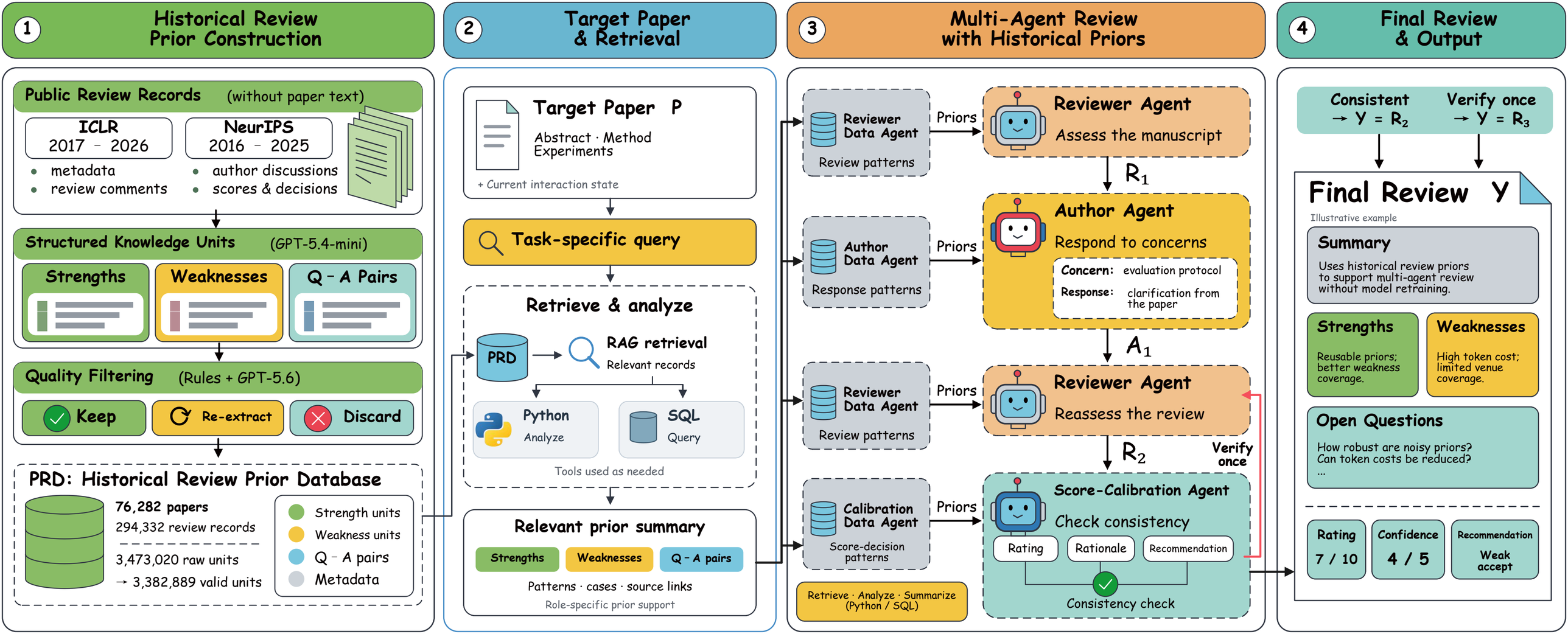

1
Historical Review Prior Construction
Public Review Records (without paper text)
ICLR 2017 – 2026
NeurIPS 2016 – 2025
metadata
review comments
author discussions
scores & decisions
Structured Knowledge Units (GPT-5.4-mini)
Strengths
Weaknesses
Q – A Pairs
Quality Filtering (Rules + GPT-5.6)
Keep
Re-extract
Discard
PRD: Historical Review Prior Database
76,282 papers
294,332 review records
3,473,020 raw units
→ 3,382,889 valid units
Strength units
Weakness units
Q – A pairs
Metadata
2
Target Paper & Retrieval
Target Paper P
Abstract · Method
Experiments
+ Current interaction state
Task-specific query
Retrieve & analyze
PRD
RAG retrieval
Relevant records
Python
Analyze
SQL
Query
Tools used as needed
Relevant prior summary
Strengths
Weaknesses
Q – A pairs
Patterns · cases · source links
Role-specific prior support
3
Multi-Agent Review with Historical Priors
Reviewer Data Agent
Review patterns
Priors
Reviewer Agent
Assess the manuscript
R1
Author Data Agent
Response patterns
Priors
Author Agent
Respond to concerns
Concern: evaluation protocol
Response: clarification from the paper
A1
Reviewer Data Agent
Review patterns
Priors
Reviewer Agent
Reassess the review
R2
Verify once
Calibration Data Agent
Score-decision patterns
Priors
Score-Calibration Agent
Check consistency
Rating
Rationale
Recommendation
Consistency check
Retrieve · Analyze · Summarize (Python / SQL)
4
Final Review & Output
Consistent → Y = R2
Verify once → Y = R3
Final Review Y
Illustrative example
Summary
Uses historical review priors to support multi-agent review without model retraining.
Strengths
Reusable priors; better weakness coverage.
Weaknesses
High token cost; limited venue coverage.
Open Questions
How robust are noisy priors?
Can token costs be reduced?
...
Rating
7 / 10
Confidence
4 / 5
Recommendation
Weak accept